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DOMAIN BREAKDOWN

code_exploration

customer_support

research_pipeline

extraction_pipeline

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code_exploration

An engineer used the agent yesterday to analyze a legacy authentication module, identifying two distinct refactoring approaches: extracting a microservice versus refactoring in-place. Today, they want to explore both approaches in depth—having the agent propose specific code changes for each—before deciding which to implement.
What's the most effective way to structure this exploration?

Your answer: B. Start two fresh sessions, manually providing a summary of yesterday's analysis findings to establish context.
Correct: D. Use 'fork_session' to create two branches from yesterday's analysis, exploring one approach in each fork.

code_exploration

An engineer who just joined the team asks the agent to help them understand the authentication and authorization architecture before making security improvements. The codebase has 800+ files across multiple services.
What exploration strategy will most effectively build understanding, given Claude built-in tools and context limits?

Your answer: C. Use Grep to find authentication entry points, read those files, then follow imports and function calls to map the auth flow incrementally.

code_exploration

Your agent has spent 25 minutes exploring a game engine's rendering subsystem—reading shader code, buffer management, and frame synchronization logic. An engineer now asks it to understand how the physics engine integrates with rendering for collision debug overlays. You notice recent responses reference "typical rendering patterns" rather than the specific VulkanPipeline and FrameGraph classes it discovered earlier.
What's the most effective approach?

Your answer: C. Summarize key rendering findings, then spawn a sub-agent for physics exploration with that summary in its initial context.

code_exploration

An agent must find why a specific error message is thrown in a large service. The most context-efficient first step is to:

Your answer: B. Search for the exact error string, then open only the files and functions that produce or handle it.

customer_support

A support agent order-status tool returns data that looks stale and contradicts what the customer sees. The agent should:

Your answer: B. Tell the customer the system shows a possibly outdated status, and verify or escalate before committing to it.

code_exploration

After adding an MCP server with specialized code refactoring tools ('extract_function', 'rename_variable', 'inline_function'), you notice the agent still uses basic text manipulation via Write and Bash sed commands for refactoring tasks. The MCP server is connected and healthy. Examining the configuration, you find each MCP tool has a minimal description like "extract_function: extracts a function from code."
What's the most effective way to improve adoption of the MCP refactoring tools?

Your answer: D. Enhance the MCP tool descriptions to explain when each tool is preferable to text manipulation and clarify expected inputs and outputs.

code_exploration

An agent is dropped into an unfamiliar repository and asked to add a feature. The best way to orient without burning context is to:

Your answer: B. Read the entry points and project structure, then search for the area the feature touches.

code_exploration

A single source file is thousands of lines long and the agent needs one function from it. The agent should:

Your answer: B. Search within the file for the function and read only that region and its immediate dependencies.

customer_support

A customer returns 4 hours after their initial session about the same billing dispute. The previous 32-turn session contains 'lookup_order' results showing "Status: PENDING, Expected resolution: 24-48 hours." In testing, you observe that when resuming sessions with stale tool results, the agent often references the outdated data in responses (e.g., "I see your refund is still being processed") even after subsequent fresh tool calls return different information.

What approach most reliably handles returning customers?

Your answer: B. Start a new session, inject a structured summary of the previous interaction (issue type, actions taken, resolution status), then make fresh tool calls before engaging.

customer_support

A frustrated customer demands a refund that the policy does not allow. The best response is to:

Your answer: B. Acknowledge the frustration, state the policy plainly, and offer the options that do exist.

customer_support

When implementing your 'lookup_order' MCP tool, the backend sometimes returns errors (e.g., "Order not found" or temporary database failures).

What is the correct pattern for communicating these errors back to the agent?

Your answer: D. Return a success response with a "status" field indicating the error type

Correct: B. Return the error message in the tool result content with the isError flag set to true

customer_support

Your agent is handling a billing dispute. After calling 'get_customer' and 'lookup_order', it identifies that the dispute involves a promotional pricing error requiring manager approval—beyond the agent's authorization level.

How should the workflow handle this mid-process escalation?

Your answer: B. Compile a structured handoff with customer details, order info, and the identified issue before calling 'escalate_to_human'.

research_pipeline

A research agent keeps spawning follow-up searches and the run is not converging. The most reliable way to prevent an endless loop is to:

Your answer: B. Give the task an explicit budget and a coverage check, and stop once the questions are answered or the budget is spent.

research_pipeline

In production, you observe that simple fact-checking queries (e.g., "What year was the Paris Climate Agreement signed?") traverse all four subagents sequentially, consuming 40+ seconds and significant tokens per query. Complex comparative research benefits from the full pipeline. Your query distribution is diverse and evolving as users discover new applications.

What's the most effective approach to optimize for varying query complexity?

Your answer: B. Create a fast-path for factual questions that bypasses subagents entirely, routing all other queries through the complete pipeline to ensure research thoroughness.

Correct: C. Have the coordinator analyze each query and dynamically decide which subagents to invoke based on its assessment of query requirements.

extraction_pipeline

A contract is too long to fit in one context window, and you need fields from across the whole document. The dependable approach is to:

Your answer: C. Summarize the document first, then extract from the summary.

Correct: B. Chunk the document with slight overlap, extract per chunk, then merge and reconcile the fields.

customer_support

An agent has tried three times to resolve a billing issue and the customer is still stuck. The right next step is to:

Your answer: B. Escalate to a human with the full history and what has been tried, so the customer does not start over.

extraction_pipeline

After implementing tool use with strict schema definitions, JSON syntax errors are eliminated, but 5% of extractions still have valid JSON with empty arrays or null values for required fields like citations and methodology. Spot-checking reveals that source documents contain this information, but in varied formats—inline citations vs. bibliographies, methodology sections vs. details embedded in introductions.

What's the most effective way to address these failures?

Your answer: B. Build a regex-based post-processing layer that scans source documents for citation patterns and methodology keywords, populating empty fields when the model fails to extract.

Correct: D. Add few-shot examples demonstrating extractions from documents with varied structures—showing how to identify citations in different formats and locate methodology details across section types.

research_pipeline

The synthesis agent receives summarized findings from the web search and document analysis agents, then passes a consolidated summary to the report generator. During testing, you discover the generated reports make factual claims without proper citations—the report generator cannot attribute statements to their original sources because that metadata was lost during the summarization steps. What's the most effective approach to ensure proper source attribution in the final reports?

Your answer: A. Have each agent output structured data separating content summaries from source metadata (URLs, document names, page numbers).

research_pipeline

After the web search agent and document analysis agent complete their tasks, the coordinator invokes the synthesis agent. However, the synthesis agent responds that it cannot complete the task because no research findings were provided. What is the most likely cause of this issue?

Your answer: B. The coordinator did not include the outputs from the previous agents in the synthesis agent's prompt.

research_pipeline

Three sub-agents searched overlapping territory and several findings repeat across their reports. Before synthesis, the coordinator should:

Your answer: C. Merge the reports, collapse duplicate findings, and keep one cited instance of each.

extraction_pipeline

Your pipeline uses a tool called 'extract_metadata' with a JSON schema for paper details. You've also defined 'lookup_citations' and 'verify_doi' tools for enrichment. During testing, you notice that when users include requests like "extract the metadata and tell me how cited it is," Claude sometimes calls 'lookup_citations' first, which fails because it needs the DOI that 'extract_metadata' would provide. What's the most effective way to ensure structured metadata extraction happens first?

Your answer: C. Set 'tool_choice' to {"type": "tool", "name": "extract_metadata"} and process the enrichment requests in subsequent turns after receiving the extracted metadata.

customer_support

A customer sends: "This is frustrating. I've explained my issue twice and nothing is being resolved. I want to talk to a real person NOW." The agent has not yet called any tools to investigate their account. What should the agent do?

Your answer: D. First call 'get_customer' and 'lookup_order' to gather account context, then escalate to a human agent.

Correct: A. Acknowledge the frustration and ask one targeted question to understand the specific issue before escalating.

research_pipeline

Production monitoring shows that follow-up queries like "summarize what we learned about market trends" consistently take 40+ seconds. Investigation reveals the coordinator spawns the synthesis subagent for each summarization request, passing 80K+ tokens of accumulated findings. The coordinator already has these findings in its context from orchestrating the research. What's the most effective way to improve response time for these follow-up summaries?

Your answer: B. Have the coordinator handle straightforward summarization requests directly using its existing context, reserving subagent spawning for complex analysis.

extraction_pipeline

An extractor pulls line items and an invoice total from a receipt. The strongest integrity check before accepting the output is to:

Your answer: B. Verify that the line items sum to the extracted total, and on a mismatch retry or flag the record.

code_exploration

An engineer asks the agent to find all callers of a function before removing it. The function is defined in a core library but is also exposed through wrapper modules that rename the function for domain-specific use (e.g., calculateTax in the library becomes computeOrderTax in the orders module).

What exploration strategy will most reliably identify all callers?

Your answer: A. Read the library and wrapper modules to identify all exposed names for the function, then Grep for each name across the codebase.

research_pipeline

The document analysis agent has a single 'analyze_document' tool that takes a document and a free-text instruction parameter. During evaluation, requests like "extract the key financial metrics" often return narrative summaries, while "summarize the methodology" sometimes returns raw data tables. The synthesis agent reports that 35% of analysis results require re-requests with clarified instructions.

What's the most effective way to improve reliability?

Your answer: A. Split the generic tool into purpose-specific tools—'extract_data_points', 'summarize_content', 'verify_claim_against_source'—each with defined input/output contracts.

code_exploration

During testing, you observe that in extended exploration sessions (30+ minutes), the agent starts giving inconsistent answers about code structure it discussed earlier. Engineers report having to repeat context about modules they've already explored. What's the most effective approach to address this?

Your answer: A. Have the agent maintain a scratchpad file that records key findings, referencing it for subsequent questions.

customer_support

A customer raises three separate issues during one session: a refund inquiry (turns 1–15), a subscription question (turns 16–30), and a payment method update (turns 31–45). At turn 48, the customer asks "What happened with my refund?" The conversation is approaching context limits.

What strategy best maintains the agent's ability to address all issues throughout the session?

Your answer: C. Summarize earlier turns into a narrative description, preserving full message history only for the active issue.

code_exploration

A README says the auth check happens in one module, but the agent must be sure before changing it. The agent should:

Your answer: B. Confirm in the current code where the auth check actually runs, then make the change there.

research_pipeline

The coordinator provides detailed step-by-step instructions to the web search subagent, specifying exact search queries, source priorities, and date filters. Production monitoring reveals three issues: (1) the subagent reports "insufficient results" rather than trying alternative approaches when pre-specified searches fail, (2) research quality drops for emerging topics that don't match expected patterns, and (3) the subagent rarely surfaces valuable tangential sources.

What's the most effective way to improve subagent adaptability?

Your answer: D. Specify research goals and quality criteria (coverage breadth, source diversity, recency) rather than procedural steps, letting the subagent determine its search strategy.

customer_support

A customer writes: "I've been going back and forth on this return for days. I just want to speak to someone who can actually help me." The agent has confirmed via 'lookup_order' that the return is straightforward—within policy and eligible for immediate processing. What should the agent do?

Your answer: A. Acknowledge frustration, inform them this is resolvable now, and offer to complete it or escalate

code_exploration

Your agent needs to insert a new helper function into the middle of a 150-line utility module, between two existing functions. The Edit tool fails because its 'old_string' parameter cannot find unique text to match — the file has repetitive docstrings, variable names, and structural patterns.

What's the most reliable way to complete this insertion?

Your answer: B. Use Edit's 'replace_all' parameter to target a common pattern and embed the new function in the replacement text

Correct: D. Use Read to load the file, add the function at the appropriate location, then Write the updated file

customer_support

After investigating a billing dispute over 25+ turns, you've identified that duplicate charges occurred due to a payment gateway timeout triggering retry logic. The required refund (\$847) exceeds your \$500 authorization limit. You need to call 'escalate_to_human', and the human agent won't have access to your conversation transcript.

What context should you pass to enable effective resolution?

Your answer: B. A structured summary: customer ID, root cause, refund amount, and recommended action.

research_pipeline

A user is expanding the research system beyond its single web search agent by adding specialized data sources. They add a financial API agent that returns structured JSON with revenue, margins, and growth rates; a news monitoring agent that returns prose summaries of recent developments; and a patent analysis agent that returns structured lists of technology areas. The synthesis agent combines these into executive briefings. Currently, it converts everything to bullet points, causing financial comparisons to lose tabular clarity and news summaries to lose narrative flow.

What change would most improve briefing quality?

Your answer: C. Update the synthesis agent to render each content type appropriately—financial data as tables, news as prose.

extraction_pipeline

Your extraction system implements automatic retries when validation fails. On each retry, the specific validation error is appended to the prompt. This retry-with-error-feedback approach resolves most failures within 2–3 attempts.

For which failure pattern would additional retries be LEAST effective?

Your answer: D. The model extracts "et al." for co-authors when the full list exists only in an external document not in the input

customer_support

Production logs reveal inconsistent error handling: when 'lookup_order' fails, the agent sometimes retries 5+ times (wasteful when the order ID doesn't exist), sometimes escalates immediately (premature for temporary network issues), and sometimes asks users for clarification (inappropriate when the issue is a backend permission error). Investigation shows your MCP tool returns uniform error

responses: [{"isError": true, "content": [{"type": "text", "text": "Operation failed"}]}. The agent cannot distinguish between error types. What's the most effective improvement?

Your answer: A. Enhance error responses with structured metadata: include `errorCategory` (transient/validation/permission), `isRetryable` boolean, and a description of what caused the failure.

research_pipeline

In production, final reports frequently contain claims without proper source attribution. Investigation shows that while the web search and document analysis agents correctly attach citations to their outputs, the synthesis agent loses track of which sources support which conclusions when combining findings.

What's the most effective architectural change?

Your answer: B. Require all subagents to output structured claim-source mappings that the synthesis agent must preserve and merge when combining findings from multiple sources.

extraction_pipeline

Your extraction pipeline processes contracts that frequently include amendments. When a contract contains both original terms and later amendments (e.g., original clause specifies "30-day payment terms" while Amendment 1 changes this to "45 days"), the model inconsistently extracts one value or the other with no indication of which applies.

What's the most effective approach to improve extraction accuracy for documents with amendments?

Your answer: A. Redesign the schema so amended fields capture multiple values, each with source location and effective date.

customer_support

You're implementing the escalation logic for when the agent should call `'escalate_to_human'`. Your team proposes four different approaches for triggering escalation.

Which approach will most reliably identify cases that genuinely require human intervention?

Your answer: B. Configure the agent to escalate after three consecutive tool calls that fail to resolve the customer's stated issue, ensuring a reasonable attempt before involving a human.

Correct: A. Instruct the agent to escalate when the customer requests a human, when the issue requires policy exceptions, or when the agent cannot make meaningful progress.

extraction_pipeline

An invoice extractor reads dates like 03/04/2025 that could be March 4 or April 3. The design that avoids silent errors is to:

Your answer: B. Require an ISO date in the output schema, and when the input is ambiguous, flag the field for review instead of guessing.

extraction_pipeline

An extractor must label each support ticket with one of five priority levels. To stop the model from inventing new labels, you should:

Your answer: B. Constrain the field to the five allowed values in the schema or tool definition, and reject anything else.

customer_support

A customer asks a simple question that the agent can answer directly from the knowledge base. The agent should:

Your answer: B. Answer the question directly and clearly, and offer escalation only if the customer needs more.

code_exploration

Your agent has analyzed a complex service module—reading 23 source files, tracing request flows, and identifying error handling patterns. A developer wants to compare two testing strategies before committing to one: end-to-end tests with mocked external services vs. snapshot tests capturing expected outputs. They need to independently develop both approaches to evaluate trade-offs. How should you manage the sessions?

Your answer: B. Resume the analysis session with `'fork_session'` enabled, creating a separate branch for each testing strategy.

code_exploration

A developer asks the agent to investigate why a specific API endpoint intermittently returns 500 errors. The codebase has 200+ files and the developer doesn't know which components are involved. The agent must trace the error through routing, middleware, business logic, and database layers.

What task decomposition approach would be most effective?

Your answer: B. Have the agent dynamically generate investigation subtasks based on what it discovers at each step, adapting its exploration plan as new information about the error path emerges.

extraction_pipeline

Your system has been operating with 100% human review for 3 months. Analysis shows that extractions with model confidence >90% have 97% accuracy overall. To reduce reviewer workload, you plan to automate high-confidence extractions. Before deploying, what validation step is most critical?

Your answer: A. Analyze accuracy by document type and field to verify high-confidence extractions perform consistently across all segments, not just in aggregate.

research_pipeline

Two sub-agents return conflicting figures for the same metric, each with moderate confidence. Before the coordinator writes the final answer, the best move is to:

Your answer: D. Include both numbers in the final answer and let the reader decide.

Correct: C. Run a focused check that re-fetches the metric from the primary source and resolves the conflict before synthesizing.

extraction_pipeline

Your extraction pipeline processes restaurant menus and must output structured JSON with fields for item names, descriptions, prices, and dietary tags. Some menus use inconsistent formatting—prices as "\$12" vs "12.00", dietary info as icons vs text.

What's the most reliable approach?

Your answer: D. Define a strict output schema and include format normalization rules in your prompt.

code_exploration

An engineer used 'Claude Code' yesterday to investigate authentication flows in a legacy monolith, building up significant context over a 2-hour session. Today she wants to continue that specific investigation. She's worked on three other codebases since then and knows the session was named "auth-deep-dive".

How should she resume?

Your answer: D. Use '--resume' auth-deep-dive to load that specific session by name

customer_support

Your 'process_refund' tool returns two types of errors: technical errors ("503 Service Unavailable", "Connection timeout") that are transient (5% of calls), and business errors ("Order exceeds 30-day return window", "Item already refunded") that are permanent (12% of calls). Monitoring shows the agent wastes 3-4 turns retrying business errors that can never succeed. Currently, both error types return only a plain text message to Claude.

What's the most effective way to reduce wasted retries while improving customer-facing response quality?

Your answer: D. Implement automatic retry logic at the tool level for technical errors only, passing business errors to Claude without retries.

Correct: A. Return structured error responses with retryable: false for business errors and a customer-friendly explanation for Claude to use.

customer_support

Compliance requires that refunds exceeding \$500 must automatically escalate to a human agent—this rule cannot be left to model discretion. Despite clear system prompt instructions, production logs show the agent occasionally processes high-value refunds directly (3% failure rate).

How should you achieve guaranteed compliance?

Your answer: C. Implement a hook to intercept tool calls; when the refund process amount exceeds \$500, block it and invoke human escalation.

code_exploration

An engineer's exploration subagent spent 30 minutes analyzing a legacy payment system, reading 47 files and documenting data flows. The session was interrupted when the engineer's connection dropped. While away, a teammate merged a PR that renamed two utility functions. The engineer wants to continue the same exploration.

What's the most effective approach?

Your answer: D. Resume the subagent from its previous transcript and inform it about the renamed functions.

extraction_pipeline

Your extraction pipeline processes invoices and extracts line items, subtotals, tax amounts, and grand totals. During evaluation, you discover that in 18% of extractions, the sum of extracted line item amounts doesn't match the extracted grand total—sometimes due to OCR errors in the source document, sometimes due to extraction mistakes by the model. Downstream accounting systems reject records with mismatched totals.

What's the most effective approach to improve extraction reliability?

Your answer: A. Add a "calculated_total" field where the model sums extracted line items alongside a "stated_total" field. Flag records for human review when values differ.

research_pipeline

When analyzing complex legal cases that cite multiple precedents, the document analysis subagent processes each sequentially. A landmark case citing 12 precedents takes over 3 minutes to analyze completely.

What's the most effective way to reduce this latency while preserving the coordinator's ability to monitor and debug the system?

Your answer: C. Have the coordinator spawn parallel document analysis subagents, each handling a subset of precedents, then aggregate results before synthesis.

extraction_pipeline

Your extraction uses tool use with a JSON schema where 'property_type' is defined as an enum: ['house', 'apartment', 'condo', 'townhouse']. After deployment, 8% of extractions fail schema validation. Investigation reveals listings mention many uncommon property types—"studio", "loft", "duplex", "mobile home", "tiny house", "converted warehouse"—and new types continue appearing regularly.

What's the most effective long-term solution?

Your answer: B. Add an "other" value to your enum with a separate 'property_type_detail' string field for specifics when "other" is selected.

extraction_pipeline

Documents arrive continuously throughout business hours and need structured data extracted. To reduce costs, you want to use the 'Message Batches API' (50% discount, up-to-24-hour processing window). Your SLA specifies that extraction results must be available within 30 hours of document arrival with 99.9% reliability.

Which batching strategy is most appropriate?

Your answer: C. Submit batches every 4 hours containing documents from that window

research_pipeline

A research agent must gather facts from eight independent web sources and produce one synthesis. None of the sources depend on each other. Which dispatch pattern stays fast without flooding the coordinator context?

Your answer: B. Dispatch eight sub-agents in parallel, each returning a short structured summary with citations, then synthesize from the summaries.

research_pipeline

When researching "renewable energy adoption," the web search agent returns recent statistics (2024: 35% adoption) while the document analysis agent extracts data from internal reports (2022: 18% adoption). The synthesis agent incorrectly flags these as contradictory sources rather than recognizing the data shows growth over time.

What change would best enable the synthesis agent to correctly interpret such temporal differences?

Your answer: A. Require subagents to include publication or data collection dates in their structured outputs.

extraction_pipeline

Your extraction system parses e-commerce product descriptions to extract specifications like dimensions, weight, and materials into JSON. Despite having a well-defined schema, the model inconsistently extracts the "materials" field—sometimes returning "cotton blend", other times "Cotton/Polyester mix", and occasionally omitting the field when material information is clearly present in the source. What's the most effective way to improve extraction consistency?

Your answer: D. Add few-shot examples showing 2-3 complete input-output pairs with standardized material description formats

extraction_pipeline

After deployment, you find that 12% of extractions contain semantic errors that pass JSON schema validation (e.g., a duration like "30 minutes" incorrectly placed in an ingredient quantity field). Human reviewers have capacity to check only 20% of extractions. Which approach most effectively allocates reviewer attention?

Your answer: A. Have the model output field-level confidence scores, then calibrate review thresholds using a labeled validation set.

code_exploration

Your codebase exploration tool stores session IDs to allow engineers to continue investigations across work sessions. An engineer spent an hour yesterday analyzing a legacy authentication module, building context about its architecture and dependencies. They want to continue today. The session ID is valid, but version control shows 3 of the 12 files the agent previously read were modified overnight by a teammate's merge.

What approach best balances efficiency and accuracy?

Your answer: C. Resume the session and inform the agent which specific files changed for targeted re-analysis

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